MADHAVAN A RAMANUJAN

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SUMMARY

Highly technical Computer Science undergraduate specializing in AI & Robotics with an 8.08 CGPA from VIT Chennai. Proven capability in end-to-end solo project architecture, standalone prototyping, and independent deployment of production-grade deep learning frameworks, complex data pipelines, and REST APIs without external technical dependencies. Certified in Oracle Cloud Infrastructure Generative AI and Cloud Architecture, with deep expertise translating complex financial analytics, forensic query heuristics, and real-time computer vision systems into scalable enterprise solutions.

EDUCATION

VIT Chennai | B.Tech, Computer Science & Engineering (AI & Robotics) | CGPA: 8.08 | *2023–2027*

TECHNICAL SKILLS

Languages: Java (Core & OOP), Python, C, C++, SQL (Advanced MySQL)
Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks
Machine Learning & Vision: TensorFlow, Keras, CNNs, Transfer Learning, OpenCV, Computer Vision, OCR, LLM APIs, Prompt Engineering
Data Analytics & Engineering: Predictive Analytics, Relational Database Design, Data Warehousing, Power BI, MS Excel, ETL Pipelines
Backend, Cloud & Tools: Flask, REST APIs, Docker, HTML/CSS, Git, GitHub, Jupyter Notebook, Arduino, Oracle Cloud Infrastructure (OCI)

CORE ENGINEERING PROJECTS

Corporate Fraud & Financial Risk Audit Pipeline *(SQL, Python, Power BI, Advanced DBMS)*
github.com/Maddy152006/Financial-Fraud-Audit-Data-Pipeline/tree/main

- Solely engineered a robust relational data warehouse database schema using MySQL to ingest and evaluate high-velocity ledger transactions against user profile entities.
- Authored forensic query detection heuristics using **CTEs** and **window functions (**`LAG`**)** to systematically isolate stolen card velocity spikes, cross-border geographic mismatches, and money laundering structuring patterns.
- Developed optimized relational database reporting views utilizing group aggregations to resolve presentation-layer data loops, cleanly exposing 880+ true risk logs to an interactive **Power BI dashboard**.

AI-Powered Dyslexia Detection System *(Python, TensorFlow, Flask, OCR, OpenCV)*
github.com/Maddy152006/dyslexia-detection-ai

- Architected a CNN-based character classification pipeline trained on 100,000+ handwritten glyph samples to detect dyslexic writing patterns across Normal, Reversal, and Corrected classes with ~96–97% validation accuracy.
- Built a Flask-based OCR inference system integrating PyTesseract, OpenCV preprocessing, adaptive thresholding, and character segmentation for real-time dyslexia screening and annotation.
- Implemented dataset augmentation, class rebalancing, confidence-based prediction logic, and confusion matrix evaluation using TensorFlow/Keras for explainable character-level analysis.

Vision-Guided Robotic Waste Segregation System *(Python, TensorFlow, OpenCV, MobileNetV2, Arduino Uno)*
github.com/Maddy152006/waste-segregation-ai

- Developed a transfer learning-based waste classification pipeline using MobileNetV2 to perform multi-class categorization across Metal, Paper/Cardboard, Plastic, and Trash classes with ~88% validation accuracy.
- Built a real-time computer vision inference system integrating OpenCV, DroidCam webcam streaming, frame preprocessing, and live prediction overlays for automated waste detection.
- Integrated TensorFlow-based prediction outputs with an Arduino Uno-controlled robotic arm via serial communication, enabling calibrated servo-based waste segregation and category-specific routing.
- Applied dataset augmentation, transfer learning fine-tuning, and confusion matrix evaluation pipelines using TensorFlow/Keras to improve classification generalization.

AI Meeting Summarizer Pipeline *(Python, Streamlit, Whisper ASR, LLMs, APIs)*
github.com/Maddy152006/meeting-summarizer

- Engineered an automated transcription and summarization web application with a modular backend architecture within a strict 3-day corporate selection challenge for Unthinkable Solution.
- Integrated Groq's Whisper-large-v3 API for highly accurate verbatim speech-to-text conversion of meeting audio.
- Leveraged a 120B parameter LLM with optimized prompts to dynamically extract executive summaries, key decisions, and structured action-item tables with deadlines.

CERTIFICATIONS

- Oracle Cloud Infrastructure Generative AI Professional
- Oracle Cloud Infrastructure AI Foundations Associate

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- **Competitive Badminton Player:** Won multiple junior badminton tournaments and competed in Karnataka State Open and state ranking tournaments.
- **Member, VIT Chennai Sports Club:** Coordinated and managed logistics and dynamic team events (2024–2025).
- **Active Participant:** Participated in the VIT Chennai 5K Run.